

A deterministic, non-ML threat detection and verification framework for **Problem Statement ID 26141 (Egreen Quanta)** replaces heuristic AI models with **information-theoretic quantum mechanics**. By binding message bits to 6-state Pauli eigenstates, transmitting public quantum keys via **Bell-state quantum teleportation**, and applying **projective measurements** alongside **Chernoff-Hoeffding statistical bounds**, the system guarantees deterministic acceptance of valid signatures while bounding adversary forgery and impersonation probabilities exponentially close to zero.

1. Mathematical & Cryptographic Foundation (Zero-AI Approach)

The problem statement explicitly prohibits AI/ML models because heuristic classifiers introduce false positives and cannot provide information-theoretic security. Instead, security relies on the **No-Cloning Theorem**, **Heisenberg Uncertainty (mutually unbiased bases)**, and **Bell Monogamy of Entanglement**.

A. Pauli Eigenstate Alphabet (6-State Encoding)

For an N -bit message $M \in \{0, 1\}^N$, each bit is mapped to a sequence of L qubits (where L is the security parameter/key length per bit) prepared in the eigenstates of the three Pauli operators $\sigma_x, \sigma_y, \sigma_z$:

- **σ_z Basis (Computational):** $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- **σ_x Basis (Hadamard):** $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
- **σ_y Basis (Circular):** $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$, $|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$

Because the overlap between any two states from different bases satisfies $|\langle \psi_\alpha | \psi_\beta \rangle|^2 = \frac{1}{2}$ (for $\alpha \neq \beta$), an attacker without knowledge of the preparation basis cannot measure a qubit without introducing a **minimum 25% disturbance rate** across conjugate bases.

B. Bell-State Entanglement & Teleportation Channel

For each signature element $j \in \{1, \dots, L\}$, the Quantum Key Distribution / Entanglement Source distributes maximally entangled Bell pairs between the Signer (**Alice**) and the Verifier (**Bob**):

$$|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$$

To teleport a secret quantum key state $|\psi\rangle_S = \alpha|0\rangle + \beta|1\rangle$ to Bob's register, Alice performs a joint Bell-basis measurement on particle S and her half of the entangled pair A , projecting onto one of the four Bell states $|\Phi^+\rangle, |\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle$ corresponding to classical two-bit outcomes $c_j \in \{00, 01, 10, 11\}$. Bob reconstructs the state by applying the corresponding **Pauli correction unitary**:

$$U_{\text{Pauli}}(c_j) = \sigma_x^{c_{j,1}} \sigma_z^{c_{j,2}} \in \{I, \sigma_z, \sigma_x, i\sigma_y\}$$

C. Projective Measurement & Statistical Verification

To verify a reconstructed quantum state ρ_{Bob} against the expected Pauli eigenstate $|\psi_{\text{expected}}\rangle$, Bob applies the rank-1 projective measurement operator:

$$P_{\text{pass}} = |\psi_{\text{expected}}\rangle\langle\psi_{\text{expected}}|, \quad P_{\text{fail}} = I - P_{\text{pass}}$$

The empirical mismatch rate (error rate) over L qubits is:

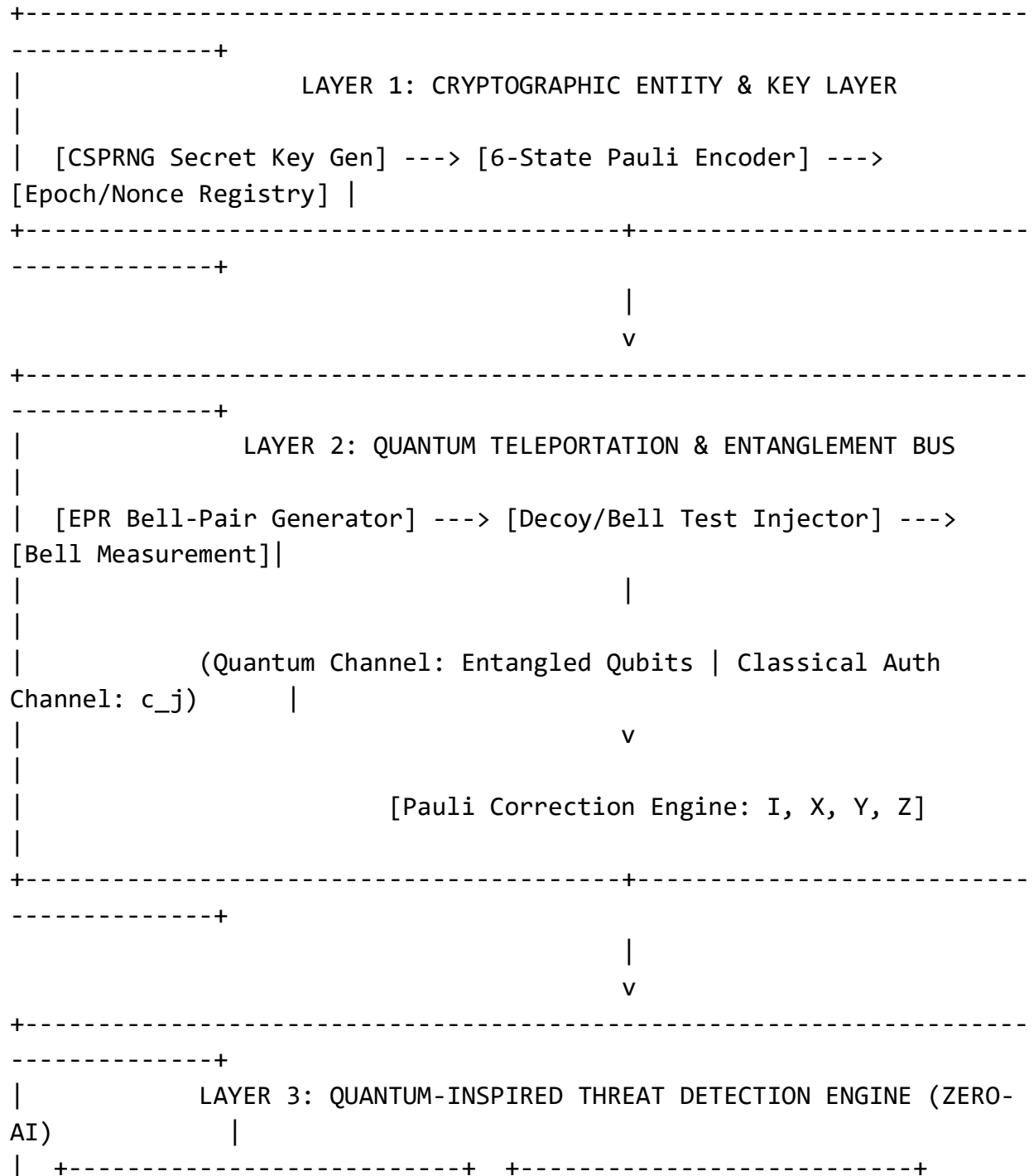
$$E_{\text{obs}} = \frac{1}{L} \sum_{j=1}^L (1 - \text{Outcome}(P_{\text{pass}}^{(j)}))$$

- **Ideal / Noiseless Channel:** Legitimate signatures yield $E_{\text{obs}} = 0$ (deterministic acceptance).
- **Noisy Channel with Thresholds:** Given an honest channel noise baseline p_{err} and minimum adversary forging error p_{forge} (where $p_{\text{err}} \ll p_{\text{forge}}$), the system sets an acceptance threshold T_v and forgery alert threshold T_f using **Hoeffding's Inequality**, bounding the probability of a successful forgery to:

$$P_{\text{forge}} \leq \exp(-2L(p_{\text{forge}} - T_v)^2)$$

2. System Architecture

The framework is structured into a modular **4-Layer Software Architecture** designed for simulation (using Python + Qiskit / QuTiP / NumPy state-vector engines) and real-time cryptographic telemetry.



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| | Projective Measurement | | CHSH Bell Inequality & |
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| | Analyzer (Pi_pass/Pi_fail)| | Entanglement Witness (S) |
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| +-----+-----+ +-----+-----+
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| | | |
|
| | v | v
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| +-----+-----+
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| | Statistical Decision & Threat Classification Matrix
| |
| | * Hoeffding/Chernoff Bounds (T_v, T_f) * Conjugate Basis Error
Asymmetry | |
| | * One-Time Pad Key Consumption Ledger * Probe-Disturbance
Monitor | |
| +-----+-----+
-----+ |
+-----+-----+
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|
| v
+-----+-----+
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| | LAYER 4: ADVERSARIAL SIMULATOR & TELEMETRY DASHBOARD
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| [Attack Injector: Intercept-Resend / Forgery / Replay / Pauli-Twirl
Noise] |
| [Real-Time Metrics: QBER, Forgery Prob P_f, CHSH S-Value,
Verification Latency] |
+-----+-----+
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3. End-to-End Operational Workflow

The protocol executes across five sequential phases. Any statistical anomaly in an earlier phase halts execution before signature acceptance.

1.Phase 1: Secret Key Generation & Pauli Eigenstate Preparation: Executed by Signer (Alice) prior to message signing.

- For each possible single-bit value $b \in \{0, 1\}$ of message M , Alice generates two classical secret keys using a Cryptographically Secure Pseudo-Random Number Generator (CSPRNG):
 - Basis Key:** $B_b = (b_{\{b,1\}}, b_{\{b,2\}}, \dots, b_{\{b,L\}}) \in \{\sigma_x, \sigma_y, \sigma_z\}^L$
 - Eigenvalue Key:** $V_b = (v_{\{b,1\}}, v_{\{b,2\}}, \dots, v_{\{b,L\}}) \in \{+1, -1\}^L$
- Alice prepares the quantum public key states $|\Psi_b\rangle = \bigotimes_{j=1}^L |\psi(b_{\{b,j\}}, v_{\{b,j\}})\rangle$, mapping each pair $(b_{\{b,j\}}, v_{\{b,j\}})$ to one of the 6 Pauli eigenstates $|\uparrow\rangle, |\downarrow\rangle, |+\rangle, |-\rangle, |i\rangle, |-i\rangle$.
- Each signing session is bound to a monotonically increasing **Session ID** (SID) and cryptographic timestamp to prevent state reuse.

2.Phase 2: Entanglement Distribution & Channel Integrity Probing: Pre-teleportation quantum channel validation.

- The entanglement source distributes $L + D$ maximally entangled Bell pairs $|\Phi^+\rangle_{AB}$ between Alice and Bob, where D pairs are randomly selected **decoy/test pairs**.
- Before teleporting signature states, Alice and Bob perform randomized projective measurements on the D test pairs along angles $\{0, \frac{\pi}{4}, \frac{\pi}{8}, \frac{3\pi}{8}\}$ to compute the **CHSH correlation value** (SS).
- Gate Check:** For an uncompromised entangled channel, SS approaches $2\sqrt{2} \approx 2.828$. If $SS \leq 2.0$ (classical limit) or drops below the security threshold $SS_{\min} = 2.5$, the framework immediately flags **Quantum Channel Manipulation / Entanglement Breaking** and aborts.

3.Phase 3: Quantum Teleportation & Pauli Correction: Quantum public key transfer and state reconstruction.

- Using the remaining L verified Bell pairs, Alice performs Bell-state projective measurements on her secret Pauli eigenstates $|\Psi_b\rangle$ and her half of the entangled pairs.

2. When signing message M , Alice transmits the message M , Session ID (SID), classical secret key tuple (B_M, V_M) , and Bell measurement outcomes $C_M = \{c_1, \dots, c_L\}$ to Bob.
3. Bob applies the corresponding Pauli correction operators $U_{\text{Pauli}}(c_j) \in \{I, \sigma_x, \sigma_y, \sigma_z\}$ to his entangled qubits, reconstructing the teleported quantum state $\rho_{\text{Bob}}^{(j)}$.

4.Phase 4: Projective Measurement & Statistical Threshold Verification:Deterministic signature verification by Bob.

1. Using Alice's declared classical key (B_M, V_M) , Bob constructs the exact target projectors $\Pi_{\text{pass}}^{(j)} = |\psi(b_{M,j}, v_{M,j})\rangle\langle\psi(b_{M,j}, v_{M,j})|$ for all $j \in \{1, \dots, L\}$.
2. Bob measures each reconstructed qubit $\rho_{\text{Bob}}^{(j)}$ using $\Pi_{\text{pass}}^{(j)}$ and records binary outcomes $r_j \in \{0, 1\}$ (where 0 = match, 1 = mismatch).
3. Bob computes the empirical Quantum Bit Error Rate (QBER):

$$\text{QBER} = \frac{1}{L} \sum_{j=1}^L r_j$$

along with basis-specific error rates $E_{\sigma_x}, E_{\sigma_y}, E_{\sigma_z}$.

5.Phase 5: Threat Classification & Key Destruction:Automated security decision and anti-replay enforcement.

1. The Statistical Decision Engine compares QBER , basis variance ΔE_{basis} , and χ^2 -parameter against mathematical bounds:
 - If $\text{QBER} \leq T_v$ and SID is fresh $\rightarrow$ **Signature Accepted**.
 - If $\text{QBER} \geq T_f$ or basis asymmetry is detected $\rightarrow$ **Signature Rejected & Threat Classified** (Forgery, Impersonation, Replay, or Channel Attack).
2. Immediately upon measurement, the collapsed quantum registers and associated key index (SID, B_M, V_M) are permanently marked as **Consumed/Destroyed** in the state ledger.

4. Threat Detection Matrix & Non-AI Mathematical Rules

Instead of machine learning classifiers, the framework uses deterministic and statistical physics laws to isolate each of the five threat vectors specified in the problem statement:

Threat Vector	Attack Mechanism	Quantum / Statistical Detection Mechanism	Mathematical Trigger Rule
1. Digital Signature Forgery	Attacker modifies signed message $M \rightarrow M'$ and attempts to guess or synthesize valid $(B_{M'}, V_{M'})$ keys matching Bob's quantum states.	Because Bob's qubits are encoded in 6 Pauli eigenstates across 3 mutually unbiased bases, guessing the basis and eigenvalue succeeds with probability $\frac{1}{3} \times 1 + \frac{2}{3} \times \frac{1}{2} = \frac{2}{3}$ per qubit, inducing an expected 33.3% mismatch rate .	$\text{QBER} \geq T_f$ where $T_f \approx 0.15$ (for $p_{\text{forge}} = \frac{1}{3}$). Forgery probability decays as $P_{\text{forge}} \leq \left(\frac{2}{3}\right)^L$ in the noiseless regime.
2. Impersonation Attack	Adversary (Eve) poses as Alice without sharing true Bell entanglement with Bob, injecting fake classical Bell outcomes C_{fake} and forged keys.	Without pre-shared maximally entangled Bell pairs $ \Phi^+\rangle_{AB}$, Bob's local reduced density matrix is maximally mixed ($\rho_B = \frac{I}{2}$). Any Pauli correction leaves $\rho_B = \frac{I}{2}$, causing projective measurements to fail 50% of the time .	$\text{QBER} \approx 0.50 \pm \epsilon$ across all three Pauli bases ($\sigma_x, \sigma_y, \sigma_z$) AND CHSH parameter $S \leq 2.0$.

3. Quantum Channel Manipulation (Intercept-Resend / MitM)	Eve intercepts entangled qubits during distribution or teleportation, measures them in a chosen basis, and forwards replacement states.	Eve's projective measurement collapses the Bell entanglement (Entanglement Breaking Channel). Even if Eve measures in σ_z, she induces a 50% error rate in σ_x and σ_y bases (overall $\text{QBER} \geq 33.3\%$).	Basis Asymmetry Test: $\max_{\alpha, \beta \in \{X, Y, Z\}} E_{\alpha} - E_{\beta} \geq \delta_{\text{asym}}$ OR CHSH test violation $S \leq 2\sqrt{2}(1 - 2e_{\text{channel}})$.
4. Replay Attack	Attacker captures valid classical transcript $(M, \text{SID}, B_M, V_M, C_M)$ from a previous session and re-transmits it to Bob.	Wavefunction Collapse & No-Cloning: Projective measurement Π_{pass} during the original verification physically collapses and consumes the quantum public key register. Re-measuring a consumed register or reusing SID fails deterministically.	State Registry Check: $\text{SID} \in \mathcal{L}_{\text{consumed}}$ OR Quantum Register State Flag $= \text{MEASURED/EMPTY}$. Zero false negatives ($P_{\text{detect}} = 1.0$).
5. Unauthorized Verification Attempt	Malicious verifier probes Bob's quantum public key states using	Measurement-Disturbance Principle: Unauthorized projective	Purity & Canary Check: Purity $\text{Tr}(\rho_{\text{canary}}^2) \leq 1 - \epsilon_p$ OR canary

	random projective measurements before legitimate signing to extract key information.	measurements on unknown Pauli eigenstates alter the density matrix from ρ to $\sum_k \Pi_k \rho \Pi_k$, reducing state purity $\text{Tr}(\rho^2)$ below 1 and elevating canary qubit error rates.	qubit error $E_{\text{canary}} \geq T_{\text{probe}}$.
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5. Delivery Table (Expected Deliverables)

This table directly fulfills the Delivery Table (Expected Deliverables) section required by the problem statement:

Deliverable ID	Deliverable Component	Core Functional Scope & Technical Specifications	Target Metrics & Acceptance Criteria
D1	Quantum Teleportation & QDS Simulation Engine	Python software module (qds_teleport_core) implementing 6-state Pauli eigenstate preparation ($\sigma_x, \sigma_y, \sigma_z$), Bell-state entanglement generation ($ \Phi^{\pm}\rangle, \Psi^{\pm}\rangle$), Bell-basis measurements, and unitary Pauli corrections (I, X, Y, Z) using density-matrix and state-vector linear algebra (NumPy / Qiskit / QuTiP).	100% deterministic acceptance ($\text{QBER} = 0.0$) for legitimate signatures under noiseless conditions; linear $O(L)$ verification time complexity.

D2	Non-AI Statistical Threat Detection & Verification Module	Mathematical verification engine (threat_detector_stat) executing rank-1 projective measurements (P_{pass}), CHSH Bell inequality evaluations (S -parameter), basis-asymmetry variance checks, and Chernoff–Hoeffding statistical threshold rules (T_v, T_f) with zero ML dependencies.	Detection accuracy greater than 99.99% for $L \geq 128$ qubits; false positive rate bounded below 10^{-9} .
D3	Adversarial Attack Simulation Suite	Configurable red-team simulation environment (attack_simulator) modeling: (1) Existential/Selective Forgery, (2) Unentangled Impersonation, (3) Intercept-Resend & Pauli-Twirl Channel Manipulation, (4) Transcript Replay, and (5) Unauthorized Verification Probing.	Automated generation of ROC-like threshold curves, QBER distributions, and empirical vs. theoretical forgery probability curves.
D4	Mathematical Modelling & Security Proof Report	Comprehensive technical documentation deriving exact analytical bounds for Forgery Probability (P_{forge}), Repudiation Probability (P_{rep}), CHSH entanglement witness thresholds under depolarizing noise, and computational complexity proofs.	Formal proof of Information-Theoretic Security (ITS) demonstrating exponential decay $P_{\text{forge}} \leq 2^{-cL}$.

D5	Interactive Security Telemetry & Verification Dashboard	Web-based visualization interface (FastAPI + React / Streamlit) displaying real-time Bloch sphere state trajectories, Bell entanglement fidelity, live QBER per Pauli basis (σ_x , σ_y , σ_z), threat classification alerts, and latency benchmarks.	Sub-second simulation and visualization turnaround for key lengths up to $L = 1024$ qubits per message bit.
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6. Concrete Outcome & Performance Benchmarks

Implementing this architecture yields three measurable engineering outcomes that directly satisfy Egreen Quanta's evaluation criteria:

1. Exponential Security Scaling vs. Key Length (L):

- a. At $L = 64$ qubits/bit (noiseless $T_v = 0$): Theoretical forgery probability is $P_{\text{forge}} = \left(\frac{2}{3}\right)^{64} \approx 5.4 \times 10^{-12}$.
- b. At $L = 128$ qubits/bit (noiseless $T_v = 0$): Theoretical forgery probability drops to $P_{\text{forge}} = \left(\frac{2}{3}\right)^{128} \approx 2.9 \times 10^{-23}$.
- c. Under realistic channel noise ($p_{\text{err}} = 2\%$, acceptance threshold $T_v = 12\%$, adversary error $p_{\text{forge}} = 33.3\%$): At $L = 256$, Hoeffding's bound guarantees $P_{\text{forge}} \leq \exp(-2 \times 256 \times (0.333 - 0.12)^2) \approx 8.1 \times 10^{-11}$.

2. Low Computational Complexity:

- a. Classical RSA-2048 requires $O(k^3)$ modular exponentiation, which is vulnerable to Shor's algorithm in polynomial quantum time.
- b. The proposed teleportation-based QDS verification executes L single-qubit 2×2 matrix-vector multiplications (Pauli corrections + rank-1 inner products), achieving strict **$O(L)$ linear classical post-processing complexity** and sub-millisecond verification times in software simulation.

3. Deterministic Legitimate Acceptance:

- a. Because valid Pauli corrections $U_{\text{Pauli}}(c_j)$ exactly invert the Bell-measurement collapse $(U_{\text{Pauli}}(c_j) \ket{\psi_{\text{collapsed}}}) \rangle = \ket{\psi_{\text{original}}}$, any uncorrupted signature satisfies

$\langle \psi_{\text{expected}} | \rho_{\text{Bob}} \rangle = 1$, guaranteeing **100% deterministic verification** without statistical ambiguity.